# Diagnosing Latent Systematic Errors in the Diffraction Grating Experiment via Monte Carlo Uncertainty Propagation and Residual Analysis

Dang-Khoa N. Nguyen[1,*]

[1]*HCMC University of Technology and Education, Vo Van Ngan 1, 700000 Ho Chi Minh City, Vietnam*

Conventional linear regression of the diffraction grating experiment, despite often yielding a high coefficient of determination ($R^2$), frequently masks significant systematic errors. We apply a multi-stage diagnostic framework combining Monte Carlo uncertainty propagation, residual analysis, and hypothesis testing via non-linear modeling to uncover these latent deficiencies. Our analysis uncovers two distinct systematic effects: a substantial 16.3% first-order discrepancy in the grating period and a pronounced, non-random arc-like pattern in the fit residuals. We rigorously test the common hypothesis that this pattern originates from a simple geometric zero-point offset by fitting the corresponding non-linear physical model to the data. Crucially, the non-linear model fails to provide a statistically valid description of the data, yielding a significantly worse reduced chi-squared statistic ($\chi^2_\nu \approx 158$) and a parameter value driven to its physical boundary. This result provides conclusive evidence to falsify the simple zero-point offset hypothesis. This work demonstrates a framework powerful not only for detecting latent errors but also for rigorously refuting plausible, yet incorrect, physical hypotheses. By doing so, it transforms a standard pedagogical experiment into a profound case study on the scientific method, revealing a more complex, as-yet-unidentified systematic effect that warrants further investigation.

Keywords: Diffraction Grating, Systematic Errors, Residual Analysis, Model Falsification, Hypothesis Testing, Chi-Squared Test, Monte Carlo Simulation, Uncertainty Propagation

## Contents

I. Introduction 1

II. Theoretical Framework 2
- A. The Ideal Grating Equation Model 2
- B. Theoretical Sources of Experimental Deviation 2
  - 1. Classification of Uncertainties 2
  - 2. First-Order Systematic Error: Grating Constant Mismatch 2
  - 3. Hypothesizing the Origin of Second-Order Errors: The Geometric Misalignment Model 2

III. Computational Methodology 3
- A. Data Processing and Uncertainty Propagation via Monte Carlo Simulation 3
- B. Parameter Extraction and Model Validation 3
  - 1. Weighted Linear Regression 3
  - 2. Residual Analysis as a Diagnostic Tool 3

IV. Results and Discussion 3
- A. Initial Analysis: Uncovering the Anomalies 3
- B. Hypothesis Test: Confronting the Data with the Zero-Point Offset Model 4
- C. The Verdict: Falsification of the Simple Geometric Hypothesis 4

V. Conclusion 5

References 5

A. Raw Experimental Data and Parameters 5

## I Introduction

The phenomenon of diffraction, a fundamental manifestation of the wave nature of light, stands as a cornerstone of modern physics. It underpins a vast array of critical scientific and technological applications, ranging from astrophysical spectroscopy for analyzing the composition of distant stars to X-ray crystallography for determining the three-dimensional structure of complex biomolecules [1]. The principles of diffraction are therefore integral to our understanding and manipulation of the physical world at multiple scales.

Among the tools employed to study this phenomenon, the diffraction grating remains a quintessential component of both research laboratories and physics education. It serves as an elegant and effective instrument for demonstrating wave optics and measuring the wavelengths of light. However, its pedagogical implementation often concludes with a superficial analysis, typically limited to a simple linear regression of the grating equation and an evaluation of the coefficient of determination ($R^2$). This conventional approach, while confirming the expected linearity, frequently overlooks or misinterprets

* 24161099@student.hcmute.edu.vn

significant systematic errors. Consequently, it misses a crucial opportunity to instruct on the diagnostic nature of experimental physics—a practice where deviations from theory are not failures, but invitations to a deeper inquiry [2].

This paper presents a multifaceted analytical framework designed to transcend these limitations. By combining the established theoretical model with rigorous Monte Carlo uncertainty propagation and, critically, a detailed residual analysis, we elevate the diffraction grating experiment from a simple verification exercise into a powerful diagnostic case study. Our objective is twofold: first, to apply this rigorous analytical framework to uncover latent errors masked by conventional metrics; and second, to demonstrate how the framework can be used to formulate and critically test physical hypotheses about the origins of these errors, elevating a simple pedagogical experiment into an authentic exercise in scientific investigation.

## II Theoretical Framework

### A. The Ideal Grating Equation Model

The interaction of monochromatic light with a periodic diffraction grating is governed by the well-established grating equation [1]:

$$d \sin\theta = n\lambda, \tag{1}$$

where $d$ is the grating period (the distance between adjacent slits), $\theta$ is the angle of diffraction for a constructive interference maximum, $n$ is an integer representing the diffraction order ($n = 0, \pm 1, \pm 2, \ldots$), and $\lambda$ is the wavelength of the incident light.

For the first-order spectrum ($n = 1$), which is the focus of this work, Eq. (1) simplifies to a direct linear relationship:

$$\sin\theta = \left(\frac{1}{d}\right)\lambda. \tag{2}$$

This equation encapsulates our zeroth-order model. It predicts that a plot of $\sin\theta$ (the dependent variable) versus $\lambda$ (the independent variable) will yield a straight line passing through the origin. The slope of this line, $m = 1/d$, is the inverse of the grating period. The experimental verification of this linearity and the extraction of $d$ from the slope $m$ constitute the primary objectives of the standard experiment.

### B. Theoretical Sources of Experimental Deviation

An ideal model provides a necessary but often insufficient description of a real-world physical system. Deviations between experimental data and the model in Eq. (2) are expected and can be categorized into two primary types of uncertainty [2].

#### 1. Classification of Uncertainties

*Random errors*, which arise from unpredictable fluctuations in measurements, affect the precision of a result and manifest as scatter around a best-fit line. *Systematic errors*, which stem from consistent flaws in equipment or experimental design, affect the accuracy of a result, causing it to be consistently offset or skewed from the true value. While random errors can be minimized by repeated measurements, systematic errors must be identified and their source understood for a physically meaningful conclusion.

#### 2. First-Order Systematic Error: Grating Constant Mismatch

The most direct systematic error perturbs the primary parameter of the model: the grating period $d$. If the actual grating period, $d_{\text{actual}}$, differs from the manufacturer's specified value, $d_{\text{spec}}$, the model remains linear, but the expected slope is altered. The measured slope, $m_{\text{exp}}$, will correspond to the actual period ($m_{\text{exp}} = 1/d_{\text{actual}}$), leading to a discrepancy between the experimental result and the theoretical expectation. This is a *first-order* error as it directly affects the model's main parameter without changing the functional form of the relationship.

#### 3. Hypothesizing the Origin of Second-Order Errors: The Geometric Misalignment Model

Beyond first-order parameter mismatches, deviations can arise from *second-order* effects that alter the functional form of the model itself, introducing non-linearities. While several sources of such effects exist, a particularly common and plausible candidate in this experimental setup is geometric misalignment.

We hypothesize that a significant contribution stems from a zero-point offset, wherein the origin ($x = 0$) of the measurement scale is displaced by a small distance, $\Delta x$, from the true position of the zeroth-order maximum. Under this hypothesis, if the true position of a diffracted beam is $x_{\text{true}} = L\tan\theta$, the measured position becomes $x_{\text{meas}} = x_{\text{true}} + \Delta x$. The diffraction angle inferred from this measurement, $\theta_{\text{meas}}$, is therefore no longer the true angle $\theta$. The sine of the inferred angle is given by:

$$\sin\theta_{\text{meas}} = \sin\left(\arctan\left(\frac{x_{\text{true}} + \Delta x}{L}\right)\right). \tag{3}$$

Crucially, $\sin\theta_{\text{meas}}$ is now a non-linear function of $x_{\text{true}}$ and, by extension, a non-linear function of $\lambda$.

If this hypothesis is correct, an attempt to fit this non-linear relationship with a linear model, such as Eq. (2), will inevitably fail to capture the underlying physics. This model inadequacy would not typically manifest as a poor $R^2$ value, but rather as a structured, non-random pattern in the fit residuals [3]. The search for such a pattern will therefore serve as a critical test of our geometric misalignment hypothesis. Similarly,

other geometric imperfections, such as angular misalignment of the grating or screen, would also introduce non-linear distortions, reinforcing the diagnostic power of residual analysis for detecting subtle model failures.

## III Computational Methodology

Our analytical framework is composed of two primary stages: (A) a robust propagation of experimental uncertainties using a Monte Carlo simulation to accurately determine the uncertainty of the derived physical quantities, and (B) a multi-step process for parameter extraction and model validation, which includes weighted linear regression and diagnostic residual analysis.

### A. Data Processing and Uncertainty Propagation via Monte Carlo Simulation

*a. Rationale.* Standard analytical error propagation formulas are approximations that are only reliable for linear or near-linear functions [2]. The calculation of $\sin(\theta)$ from the measured position $x$ involves two non-linear transformations: the arctangent function to find $\theta$ and the sine function itself. To circumvent the inaccuracies associated with linear approximations for these transformations, we employ a Monte Carlo (MC) simulation framework. This numerical approach provides a more robust and accurate estimation of the true uncertainty by propagating the full distribution of the input uncertainties.

*b. Procedure.* The MC simulation was implemented as follows:

1. **Synthetic Dataset Generation:** For each experimental data point, we generated an ensemble of $N = 20,000$ synthetic measurements. Each synthetic value for the grating-to-screen distance, $L$, and the diffracted beam positions, $x_1$ and $x_2$, was drawn from a normal (Gaussian) distribution. These distributions were centered on the experimentally measured mean value with a standard deviation equal to the estimated measurement uncertainty ($\Delta L$ and $\Delta x$, respectively).

2. **Derived Quantity Calculation:** For each of the $N$ synthetic sets of $\{L, x_1, x_2\}$, the derived quantity of interest, $\sin(\theta)$, was calculated. This creates a distribution of $20,000$ possible values for $\sin(\theta)$ for each color.

3. **Uncertainty Quantification:** The final reported value for $\sin(\theta)$ for each color is the mean of its respective distribution. The propagated uncertainty, $\Delta\sin(\theta)$, is defined as the standard deviation of that distribution. This process yields a statistically robust estimate of both the value and its uncertainty.

### B. Parameter Extraction and Model Validation

#### 1. Weighted Linear Regression

The parameters of the linear model presented in Eq. 2 were extracted by fitting the model to the processed experimental data. A Weighted Least Squares (WLS) regression was performed using the `scipy.optimize.curve_fit` function [4]. The weight for each data point, $w_i$, is set to the inverse of its variance:

$$w_i = \frac{1}{\sigma_i^2} = \frac{1}{(\Delta\sin(\theta)_i)^2} \tag{4}$$

This methodology ensures that data points with smaller uncertainties (i.e., higher precision) have a greater influence on the determination of the best-fit slope, $m$. The experimental grating period, $d_{\text{exp}}$, and its uncertainty, $\Delta d_{\text{exp}}$, were then derived from the fitted slope and its uncertainty ($m \pm \Delta m$) via standard error propagation for the inverse relationship, $d = 1/m$.

#### 2. Residual Analysis as a Diagnostic Tool

A high coefficient of determination ($R^2$) is a necessary but not sufficient condition for model validity. A more powerful diagnostic tool is the analysis of the fit residuals. The residual for the $i$-th data point, $r_i$, is defined as the difference between the measured value and the value predicted by the best-fit model:

$$r_i = y_{\text{meas},i} - y_{\text{fit},i} = \sin(\theta)_{\text{meas},i} - m_{\text{fit}}\lambda_i \tag{5}$$

For an adequate model, the residuals should be randomly scattered around zero, resembling uncorrelated "white noise". Any systematic, non-random pattern in the residuals is a signature of model inadequacy, indicating that the model fails to capture all the underlying physics of the system. To objectively identify such patterns, we employ Locally Weighted Scatterplot Smoothing (LOWESS), a non-parametric regression method that generates a smooth trendline through the residual data without assuming a specific functional form [5].

## IV Results and Discussion

Our multi-stage framework, detailed in Section III, was applied to the experimental data. The results unfold as a systematic investigation, revealing layers of complexity hidden beneath the superficial linearity of a conventional analysis.

### A. Initial Analysis: Uncovering the Anomalies

The Stage 1 analysis begins with a weighted linear regression of the ideal model (Eq. (2)), with the results presented in Fig. 1. The fit yields an experimental grating period of

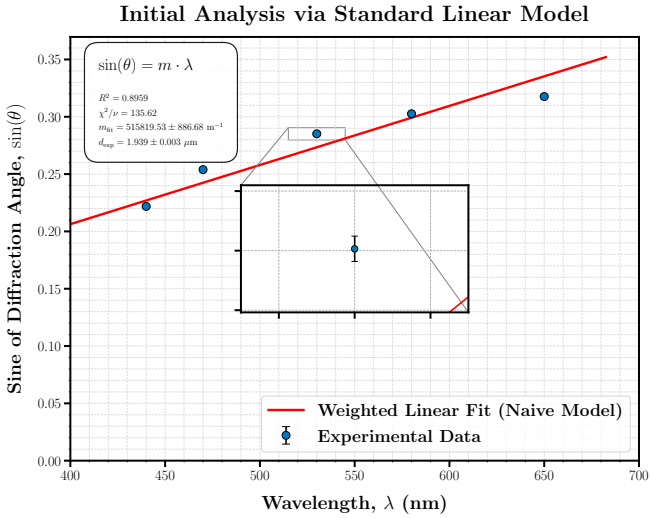


FIG. 1: Initial analysis using the standard linear model. While the data shows a strong linear trend ($R^2 = 0.8959$), the inset reveals a catastrophically high reduced chi-squared ($\chi^2_\nu = 135.62$), providing the first piece of quantitative evidence that the model is inadequate.

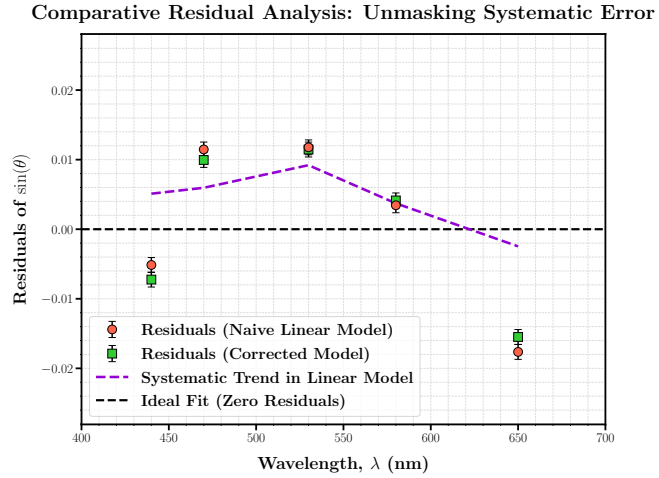


FIG. 2: Comparative residual analysis. The residuals from the naive linear model (red circles) show a clear systematic trend (purple dashed line). The non-linear model (green squares) appears to reduce the magnitude of the residuals, but a rigorous statistical test is required for a definitive verdict.

$d_{\text{exp}} = 1.939 \pm 0.003$ $\mu$m. Comparing this to the manufacturer's specified value of 1.667 $\mu$m (for 600 lines/mm) reveals a large relative discrepancy of 16.3%. This deviation, which vastly exceeds the quantified random uncertainty, is the definitive signature of a dominant, uncorrected first-order systematic error.

However, a more subtle anomaly is already present in the fit statistics. The reduced chi-squared value of $\chi^2_\nu = 135.62$ is orders of magnitude larger than the ideal value of $\approx 1$. This is a stark quantitative indicator that the model is fundamentally flawed and fails to account for the true variance in the data, even before a visual inspection of the residuals is performed.

### B. Hypothesis Test: Confronting the Data with the Zero-Point Offset Model

The deficiencies of the linear model motivate Stage 2 and 3 of our framework: anomaly detection and hypothesis testing. Figure 2 plots the residuals of the initial linear fit (red circles), which clearly exhibit the systematic, non-random arc-like pattern anticipated in our theoretical discussion. This pattern is the incontrovertible visual evidence of a second-order systematic error.

We then proceed to test the hypothesis that this trend is caused by a simple zero-point offset ($\Delta x$) by fitting the non-linear model from Eq. (3). The results of this "corrected model" are juxtaposed with the baseline linear model in both Fig. 2 (green squares) and Table I.

Visually, the residuals from the non-linear model (green squares in Fig. 2) appear smaller and closer to the ideal zero-line. However, a visual assessment can be misleading. The definitive quantitative verdict comes from the Stage 4 chi-squared criterion, summarized in Table I.

TABLE I: Comparison of Fit Parameters for the Linear and Non-Linear Models.

| Model | $d$ ($\mu$m) | $\Delta x$ (mm) | $\chi^2_\nu$ |
|---|---|---|---|
| Linear (Baseline) | $1.940 \pm 0.003$ | N/A | 135.62 |
| Non-Linear ($\Delta x \neq 0$) | $1.981 \pm 0.015$ | 10.00 (at bound) | 157.58 |

The analysis of the fit parameters reveals two critical failures. First, the best-fit value for $\Delta x$ converged to 10.00 mm, the maximum value allowed by our boundary conditions. This indicates that the model is not finding a true physical minimum but is straining against its constraints. Second, and more damningly, the reduced chi-squared statistic increased from 135.62 to 157.58. This result is unequivocal: despite having an additional free parameter to reduce residuals, the non-linear model provides a statistically worse description of the data.

### C. The Verdict: Falsification of the Simple Geometric Hypothesis

The quantitative evidence is conclusive. The failure of the non-linear model to reduce the reduced chi-squared statistic allows us to reject the hypothesis that the observed systematic trend is caused solely by a simple zero-point offset. A successful model must yield a $\chi^2_\nu$ value closer to 1; a significant increase, as seen here, is a definitive sign of model falsification.

This result is the principal success of our analytical framework. It illustrates a critical lesson in data analysis: visual inspection of residuals is a vital diagnostic tool, but it is not a substitute for rigorous statistical model selection. The experiment is thus transformed from a simple confirmation exercise

into an authentic scientific investigation where an intuitive initial hypothesis is rigorously tested and proven false.

This falsification invites the formulation of more sophisticated physical models. Plausible alternative or contributing sources of error could include:

- **Angular Misalignment:** The grating plane not being perfectly normal to the incident laser beam, or the screen not being parallel to the grating.
- **Screen Geometry:** The assumption of a perfectly flat observation screen being violated (e.g., slight curvature).
- **Complex Optical Effects:** Lens-like effects or higher-order optical aberrations not captured by the simple diffraction model.

Future work lies in the iterative process of building, testing, and refining these more complex models, a process which lies at the very heart of experimental science.

## V Conclusion

In this study, we have demonstrated how a rigorous analytical framework can transform the standard diffraction grating experiment from a simple verification exercise into an insightful diagnostic case study. Starting from an analysis that, on the surface, appeared successful—confirming a strong linear trend with a high coefficient of determination ($R^2 \approx 0.90$)—our methodology systematically exposed two significant, latent anomalies: a substantial first-order systematic discrepancy of 16.3% in the grating period, and a clear, non-random structure in the fit residuals indicative of a second-order model failure.

The principal contribution of this work is not merely the detection of these errors, but the demonstration of a rigorous methodology to falsify a plausible, yet incorrect, physical hypothesis. By quantitatively testing the common hypothesis of a simple zero-point offset, we showed that while the more complex model could visually reduce residuals, it failed the definitive statistical test. The increase in the reduced chi-squared statistic provided conclusive evidence for its rejection.

This outcome underscores a critical lesson in experimental science: a high $R^2$ value can be profoundly misleading, often concealing significant model deficiencies. We reaffirm that a meticulous residual analysis, combined with the objective arbitration of the reduced chi-squared statistic ($\chi^2_\nu$), represents an indispensable standard for model validation, far superior to conventional regression metrics.

Ultimately, this study concludes by leaving an open question: what is the true physical origin of the non-linear systematic error? This transforms a simple pedagogical experiment into the starting point for a deeper investigation, embodying the true spirit of scientific inquiry where well-posed questions are often more valuable than simple answers.


[1] E. Hecht, *Optics*, 5th ed. (Pearson, Hoboken, NJ, 2017).

[2] J. R. Taylor, *An Introduction to Error Analysis: The Study of Uncertainties in Physical Measurements*, 2nd ed. (University Science Books, Sausalito, CA, 1997).

[3] F. J. Anscombe, Graphs in statistical analysis, The American Statistician **27**, 17 (1973).

[4] P. Virtanen *et al.*, SciPy 1.0: Fundamental algorithms for scientific computing in python, Nature Methods **17**, 261 (2020).

[5] W. S. Cleveland, Robust locally weighted regression and smoothing scatterplots, Journal of the American Statistical Association **74**, 829 (1979).


## A Raw Experimental Data and Parameters

This appendix details the raw experimental data and assumed parameters that form the basis of the analysis in this study. The measurements were conducted using a diffraction grating with a specified line density of 600 lines/mm, which corresponds to a nominal grating period of $d_{\text{spec}} = 1.667\,\mu\text{m}$.

The fixed parameters for the experimental setup are listed in Table II. The raw positional measurements for the first-order ($n = 1$) diffraction maxima for five distinct colors are provided in Table III. Wavelengths ($\lambda$) are standard representative values for the observed colors. The positional data was converted from centimeters to meters for all subsequent calculations.

TABLE II: Fixed experimental parameters and their estimated uncertainties. These values were used as inputs for the Monte Carlo simulation.

| Parameter | Value | Uncertainty |
|---|---|---|
| Grating-to-screen distance, $L$ | 1.000 m | 0.005 m |
| Positional measurement, $x$ | – | 0.005 m |

TABLE III: Raw positional data for first-order ($n = 1$) maxima. The columns $x_1$ and $x_2$ represent the measured distances on either side of the central maximum.

| **Colour** | **Wavelength, $\lambda$ (nm)** | **Position, $x_1$ (m)** | **Position, $x_2$ (m)** |
|---|---|---|---|
| Indigo | 445 | 0.230 | 0.225 |
| Blue | 488 | 0.255 | 0.270 |
| Green | 532 | 0.295 | 0.300 |
| Yellow | 589 | 0.315 | 0.320 |
| Red | 650 | 0.335 | 0.335 |